
Effectiveness of AI-enhanced virtual patients in clinical reasoning training of healthcare professionals and students: a systematic review

Received: 4 June 2026

Accepted: 20 August 2026

Published online: 24 August 2026

Cite this article as: Mofidi S.A. & Khoshgoftar Z. Effectiveness of AI-enhanced virtual patients in clinical reasoning training of healthcare professionals and students: a systematic review. *BMC Med Educ* (2026). <https://doi.org/10.1186/s12909-026-10242-6>

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Title Page

Title

Effectiveness of AI-Enhanced Virtual Patients in Clinical Reasoning Training of Healthcare Professionals and Students: A Systematic Review

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Abstract

Background: Clinical reasoning is a core competency for healthcare students and professionals. However, current educational approaches may not adequately prepare learners for its complexity in practice. Virtual patients (VPs) provide a structured strategy for clinical reasoning education, and artificial intelligence (AI) may enhance their interactivity, adaptability, and feedback capacity.

Aim: This study aimed to assess the effectiveness of AI-enhanced VPs in clinical reasoning training among healthcare professionals and students.

Methods: We systematically searched PubMed/MEDLINE, Scopus, and ERIC. Records were screened at title/abstract and full-text stages by two independent reviewers. Data were extracted using a standardized form. Findings were synthesized descriptively, and methodological quality was assessed using the Mixed Methods Appraisal Tool (MMAT) 2018.

Results: The search retrieved 921 records, and 16 studies were included, with a total of 1,136 participants. Overall, 10 studies (63%) favored AI, 2 (13%) reported mixed findings, and 4 (25%) were not applicable. The most promising direction of effects was observed for clinical reasoning and history-taking performance, with 7 of 8 studies (88%) favoring AI. Favorable findings were also reported for learner perception or satisfaction (3/5, 60%), clinical competence/skills (3/5, 60%), diagnostic accuracy (1/3, 33%), and knowledge/retention (3/3, 100%). By AI role, tutor/mentor systems favored AI in 5 of 6 studies (83%), feedback generators in 8 of 10 (80%), simulation-only in 1 of 5 (20%), and adaptive narratives in 1 study.

Conclusion: AI-enhanced VPs show promising potential for supporting clinical reasoning education, especially when used for history-taking, reasoning practice, feedback, or tutor/mentor support.

Keywords:

Simulation Training; Artificial Intelligence; Clinical Decision-Making; Medical Education;
Large Language Models

Background

Clinical reasoning (CR) is a fundamental competency in healthcare practice. It can be described as a context-sensitive cognitive process that guides professional judgment and decision-making in clinical settings (1). This process involves integrating clinical information, formulating and testing hypotheses, interpreting data, and making management decisions within complex and evolving environments. Strong clinical reasoning abilities are directly linked to patient safety, as accurate diagnostic and therapeutic decisions depend on well-developed reasoning skills (2).

Despite its importance, structured, evidence-based approaches for explicitly teaching and assessing clinical reasoning remain insufficiently integrated into many medical and health professions curricula (3). Concerns persist regarding gaps in critical thinking and reasoning skills among nursing students and newly graduated healthcare professionals, suggesting that current educational strategies may not fully prepare learners for real-world clinical challenges (4). Conventional instructional methods—such as lectures, case-based discussions, and opportunistic bedside learning—often fail to reflect the dynamic, uncertain, and multifactorial nature of authentic clinical decision-making (5).

To address these limitations, virtual patients (VPs) have been developed as technology-enhanced educational tools that simulate clinical encounters. Virtual patients are generally defined as computer-based simulations that replicate real-life patient scenarios, enabling learners to interact

with clinical cases in a structured and safe environment (6, 7). First introduced in the early 2000s in response to increasing educational demands and limited clinical training opportunities, VPs have increasingly been integrated into health professions education (7). Compared to standardized patients, VPs provide advantages such as reproducibility, scalability, flexibility, and freedom from issues like fatigue or inconsistent performance (8). They are increasingly used as adjuncts to real clinical experiences and are broadly accepted as safe platforms for developing clinical reasoning competencies (9).

Empirical findings indicate that VP-based learning can enhance various dimensions of clinical reasoning, including information gathering and problem-solving abilities (10). Interactive and conversational VP systems have been associated with improved clinical competencies, particularly when learners perceive the interactions as realistic and engaging (7). However, traditional computer-based VP implementations may lack authenticity and depth, which can reduce their educational effectiveness (9). Moreover, many existing VP platforms primarily target measurable components of reasoning while offering limited support for more advanced processes, such as constructing differential diagnoses, refining hypotheses, and articulating diagnostic justification (11). Consequently, although VPs represent a meaningful innovation in simulation-based education, further refinement is necessary to maximize their impact on higher-order reasoning skills.

Recent developments in artificial intelligence (AI), particularly the emergence of large language models (LLMs), have introduced new opportunities to enhance VP systems. LLMs show promise in advancing medical education through interactive simulations, adaptive guidance, and personalized feedback (12). By leveraging AI-driven natural language processing and adaptive

algorithms, VP platforms can now generate more complex and contextually rich clinical scenarios, offering responses that more closely resemble real patient interactions (13). The incorporation of LLMs into VP environments has been proposed as a strategy to increase scenario diversity, learner engagement, and individualized feedback while better simulating the uncertainty inherent in real-world clinical reasoning (14). Additionally, future integration with immersive technologies such as virtual reality and haptic systems may further strengthen the realism and experiential quality of these simulations (8).

Given the essential role of clinical reasoning in ensuring safe and effective patient care, and considering the rapid technological advancements in AI-enhanced virtual simulations, it is important to systematically examine the current evidence base. This systematic review aims to evaluate the effectiveness and other descriptive education outcomes of AI-enhanced virtual patients in improving clinical reasoning skills among health professionals and trainees.

Methods

Study design

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (15). The protocol was developed a priori to define the research question, search strategy, eligibility criteria, and methods for data extraction and synthesis. The review protocol was registered in PROSPERO (PROSPERO ID: CRD420261388649).

Eligibility criteria

Studies were included if they met all of the following criteria:

1. **Population:** Undergraduate or postgraduate health professions students or practitioners, including medical, nursing, pharmacy, and allied health disciplines.
2. **Intervention:** AI-powered virtual patients designed to enhance clinical reasoning skills.
AI-powered virtual patients were defined as interactive digital simulations incorporating adaptive algorithms, natural language processing, machine learning, or automated feedback systems. Additional file 1 offers the operational definition of AI-VPs used in this criteria.
3. **Comparator:** Any of the following: traditional teaching methods (e.g., lectures, paper-based cases), non-AI virtual patients, standardized patients, or no intervention.
4. **Outcomes:** Any measure related to clinical reasoning, including but not limited to: diagnostic accuracy, clinical decision-making scores, script concordance test performance, problem-solving ability, or validated clinical reasoning assessment tools.
5. **Study Design:** Randomized controlled trials, quasi-experimental studies, controlled before–after studies, cohort studies, cross-over studies, or observational studies.

Studies were excluded if they were not published in English, conference abstracts without full texts, commentaries, editorials, or purely descriptive technical reports without educational outcomes of clinical reasoning.

Search Strategy

A comprehensive search was conducted in PubMed/MEDLINE, Scopus, and ERIC from inception to 29 April 2026. The search combined controlled vocabulary (e.g., MeSH terms) and free-text keywords for three main concepts:

1. Artificial intelligence
2. Virtual patients
3. Clinical reasoning

Synonyms for each concept were combined with OR, and the three main concepts were combined with AND. The full search strategy for each database is provided in Additional file 2.

Study selection

All retrieved records were imported into a reference management software (EndNote 21), and duplicates were removed. Two independent reviewers screened titles and abstracts for eligibility. Full texts of potentially relevant articles were then assessed independently by the same reviewers. Disagreements were resolved through discussion.

Data extraction

A standardized data extraction form was developed and pilot-tested. Extracted variables included:

- Study identifiers: author(s), year, country, setting
- Study design category

- Study aim
- Participant characteristics: discipline, level of training, sample size, mean age
- Intervention characteristics: specialty of cases, type of AI used, duration, presence of AI feedback, primary AI role
- Comparator characteristics: type and description
- Outcome measures: domain, measurement tool, scoring or assessment method
- Main findings: direction of effects, summary of results
- Duration and follow-up period

Extraction was conducted independently by two reviewers, with discrepancies resolved through discussion.

Quality assessment

Methodological quality was assessed using the Mixed Methods Appraisal Tool (MMAT) 2018, appropriate for heterogeneous study designs, including randomized controlled trials, non-randomized studies, pre-post studies, quantitative descriptive studies, and mixed-methods studies (16). This tool is both evaluated and validated by Hong et al. (17, 18). Utilizing this tool includes 4 steps: applying the screening questions, determining the study design, carrying out the appraisal, and concluding on the overall quality of each study. Two reviewers independently performed quality assessment. Any disagreements were resolved through discussion. No overall numerical quality scores were calculated; criterion-level ratings were recorded for transparency. In order to assess the descriptive methodology and quantitative methodology of mixed-methods studies separately, we also assessed each mixed-methods study against the descriptive criterion

and the relevant quantitative criterion (Randomized or non-randomized dependent on the study design).

Data synthesis

A narrative synthesis was conducted due to heterogeneity in study designs, interventions, comparators, and outcome measures. No meta-analysis was performed. Following steps were performed for a systematic narrative synthesis:

1. **Coding the characteristics of studies:** each study characteristic was classified into predefined categories. Study design was coded into randomized controlled studies, non-randomized controlled studies, pre-post comparison studies, descriptive, validation, feasibility, observational studies, and cross-over studies. Classification was also carried out based on the type of VP used in each study. These contained: chatbot-style virtual patients, mixed-reality simulation, social robotic virtual patient, intelligent tutoring simulation systems, and case-simulation platforms. Another characteristic was the AI role in designing VPs which was categorized into these groups; simulation-only, AI tutor/mentor, and adaptive narrative systems. AI-role categorization was not exclusive and one study could have several AI-roles and as a result, be categorized in multiple AI-role categories. We also grouped studies based on whether they used any form of AI assisted feedback in their VP design or not. Other factors were having a comparator and intervention dosage and outcome domains of clinical reasoning. In order to improve the reproducibility of this review exact definition of each category is provided in Additional File 1.

2. **Coding the overall direction of effects of each study:** each study overall direction of effects was coded to determine if results were in favor of AI, in favor of the comparison group, mixed, or not applicable. Only if a study provides the statistical significance of the direction of effects (i.e. p-values), it was considered to be in favor of AI or comparison. Thus, if it failed to do so, it was categorized as not applicable. If in a study some of the outcomes of the study favored AI and some favored the comparator; the direction of effects of the primary outcome was used to demonstrate the overall direction of effects. If a study did not have any form of comparator, it was marked not applicable as it couldn't provide any interpretation of the statistical significance. If a study reported some outcomes to be in favor of comparison and some in favor of AI, the results were determined to be mixed.
 3. **Assessing the overall direction of effects based on characteristics:** we tried to answer the different overall direction of effects in studies by studying it in each group of outcome domain, AI role in VP design, Feedback, and study design.
 4. **Coding the outcome-level direction of effects:** To assess the direction of effects for individual outcomes independently of other outcomes reported within the same study, each outcome was coded separately. The direction of effect for each outcome was classified using the same criteria applied to the overall study-level direction of effects.
- Thus, descriptive analyses addressed three main questions:
- **Overall study-level direction of effects:** Count and percentage of studies favoring AI, mixed, no difference, comparator, or not applicable.

- **Direction by outcome domain:** Studies were grouped by predefined clinical reasoning domains. Counts and percentages of studies favoring AI, mixed, no difference, comparator, or not applicable were calculated.
- **Direction by AI role and feedback presence:** Studies were categorized by primary AI role (simulated patient, tutor/mentor, feedback generator, adaptive narrative) and by presence or absence of AI feedback. Counts and percentages for each stratum were summarized.
- **Outcome-level direction of effects:** The direction of effect for each outcome within each study was assessed independently of the other outcomes reported in that study.

Results

Search and Screening

The study selection process is summarized in the PRISMA flow diagram (Figure 1). A total of 921 records were identified through the search process. After removal of 166 duplicate records, 755 records remained and entered title and abstract screening.

During primary screening, 680 records were excluded. The reasons for exclusion were: wrong study design ($n = 190$), no education component ($n = 63$), no AI component ($n = 111$), no virtual patient component ($n = 61$), participants were not healthcare providers or healthcare learners ($n = 42$), and no clinical reasoning outcome or focus ($n = 213$).

Following primary screening, 75 reports proceeded to full-text screening. Full text of 5 studies was not found and 70 reports entered the eligibility assessment. At this stage, 54 reports were excluded. The main reasons for full-text exclusion were wrong type of intervention ($n = 16$) and

desired outcomes not reported ($n = 38$). Overall, 16 studies met the eligibility criteria and were included in the final synthesis.

General Characteristics of Included Studies

Sixteen studies were included in the final synthesis. Six studies were randomized controlled trials (19-24), three were non-randomized controlled or comparative studies (25-27), two used pre-post self-comparison designs (28, 29), four were descriptive, validation, feasibility, or observational studies (30-33), and one used a cross-sectional crossover design (34). Three reports explicitly included mixed-methods components (20, 23, 33) (Table 1).

Across the included studies, the extracted sample sizes summed to 1,136 participants. Individual study sample sizes ranged from 15 participants (28) to 210 participants (31), with a median sample size of 51 and a mean sample size of 71 (Table 1).

Most studies enrolled students or undergraduate learners, including medical students (19, 24-26, 28, 30-33), nursing students (20, 23), Korean medicine students (22), and dental students (34). Two studies enrolled postgraduate residents, including obstetrics and gynecology residents (29) and breast oncology residents (21). One study enrolled practicing primary care providers participating in continuing medical education (27) (Table 1).

Characteristics of AI-Enhanced Virtual Patient Interventions

The included studies used a range of AI-enhanced virtual patient formats. 11/16 interventions were delivered through text-based or chatbot-style virtual patients (19, 20, 22-25, 28, 31-34).

Less commonly, studies used more complex simulation environments, including mixed-reality simulation with an AI mentor (21), social robotic virtual patient cases (26), online or intelligent tutoring simulation systems (27, 30). One study used a case simulation platform (29) (Table 2).

The role of AI differed across interventions. In several studies, AI primarily acted as the simulated patient, supporting learner interaction with a clinical case or patient dialogue without having any other roles (5/16 studies; 31%). In other studies, AI had a more instructional role, functioning as a tutor/mentor (6/16 studies; 38%). One study used AI as an adaptive narrative system in designing VPs (Table 2).

Feedback was a common but not universal feature. Several interventions included automated or AI-supported feedback, usually related to learner performance, reasoning, interviewing, or diagnostic errors (10/16 studies; 63%) (19, 21, 22, 24-30). Other studies focused mainly on the virtual patient interaction itself and did not clearly include AI-generated feedback as a separate intervention component (6/16 studies; 37%) (20, 23, 31-34) (Table 2).

Intervention dose varied substantially. Some studies evaluated a single session or single virtual patient interaction (20, 22, 32, 33). In contrast, others used multi-session or longer training formats, ranging from several cases or repeated sessions to multi-week or rotation-based training (19, 21, 24, 26, 28, 29, 31). In several studies, the intervention duration was not clearly reported or was not presented as a fixed training period (23, 25, 27, 30, 34) (Table 2).

The clinical domains were also diverse. Many studies focused on history taking, interviewing, diagnostic reasoning, or general clinical decision-making (19, 24, 25, 28, 30, 31). Others were situated in more specific clinical areas, including nursing diagnostic reasoning (20, 23), obstetrics and gynecology (29), depression and suicidality assessment (32), breast oncology (21), Korean medicine OSCE preparation (22), case management (33), orofacial pain and temporomandibular disorders (34), and rheumatology or axial spondyloarthritis (26, 27) (Table 2).

Comparator Conditions

The comparator conditions varied across studies. Ten studies included an external comparator group (62.5%) (19-27, 34). Two studies used participants' own pre-intervention performance as the comparison point (12.5%) (28, 29), while four studies had no comparator condition (25.0%) (30-33) (Table 2).

A variety of comparators were used across the included studies. Two studies compared AI-enhanced virtual patient training with the same or similar virtual patient experience without AI feedback (19, 26). Three studies used traditional or conventional training controls (21, 23, 25). One study used a static, non-AI case narrative as the comparator (20), and one compared the intervention with traditional role-play (24). Other comparator formats included peer role-play (22), a standardized real-patient encounter (34), and live expert-led continuing medical education (27) (Table 2).

Outcome Domains

The included studies assessed a range of educational and clinical reasoning outcomes. At the study level, the most common outcome area was clinical reasoning or history-taking performance, assessed in 8/16 studies (50.0%) (19, 20, 22-26, 29). Clinical competence or procedural/clinical skills were assessed in 5/16 studies (31.3%) (21, 22, 26, 29, 33), and learner perception, satisfaction, or usability was assessed in 5/16 studies (31.3%) (20, 22-24, 32) (Table 3).

Less frequently assessed outcomes included diagnostic accuracy, reported in 3/16 studies (18.8%) (27, 32, 34), 3 clinical reasoning process measures, reported in 3/16 studies (18.8%) (30, 31, 33), knowledge or retention, reported in 3/16 studies (18.8%) (21, 28, 29), and

communication or interaction outcomes, reported in 3/16 studies (18.8%) (22, 23, 34). Feasibility or AI fidelity outcomes were reported in 2/16 studies (12.5%) (25, 32) (Table 3).

Measurement approaches were heterogeneous. Most studies used performance-based assessments, including rubrics, checklists, rating scales, standardized assessments, or OSCE-based tools (19-26, 29, 33). Other studies used MCQs, case-based assessments, diagnostic tasks, self-report questionnaires, surveys, interviews, or process-coding approaches, depending on the outcome being measured (27, 28, 30-32, 34) (Table 3).

Quality Assessment

Quality assessment of studies is summarized in Table 4. Methodological quality varied across the included studies. In randomized controlled trials, the main recurring concern was incomplete reporting of the randomization process, including sequence generation or allocation concealment. In non-randomized studies, the main concerns were limited control for confounding and unclear participant representativeness. In descriptive studies, sample representativeness and selection/nonresponse bias were the most frequent issues. For mixed-methods studies, the main concern was incomplete clarity about whether both qualitative and quantitative components fully met the relevant methodological criteria. Detailed criterion-level ratings are presented in Additional file 3.

Direction of Effects

At the study level, the majority of included studies reported outcomes in favor of AI (Table 3). Out of 16 studies, 10 (63%) favored AI, 2 (13%) were mixed, and 4 (25%) were not applicable. Looking across outcome domains (Figure 2), the strongest support was seen for clinical reasoning and history-taking performance, with 7 of 8 studies (88%) favoring AI. For learner perception or satisfaction, 3 of 5 studies (60%) favored AI, and for clinical competence/skills, 3

of 5 studies (60%) favored AI. Diagnostic accuracy was supported in 1 of 3 studies (33%) and all studies measuring knowledge/retention (3/3, 100%) favored AI. Other domains, including clinical reasoning/process coding (0/3) and feasibility/AI fidelity (1/2), were mostly not applicable or descriptive.

Considering feedback presence, among 10 studies that included AI-generated feedback, 8 (80%) favored AI, 1 was not applicable (10%), and 1 (10%) was mixed (Figure 3). In contrast, among 6 studies without feedback, 2 (33%) favored AI, 1 (17%) was mixed, and 3 (50%) were not applicable.

Direction of effects also varied by study design and AI role (Figure 3). Among randomized controlled trials, 5 of 6 studies (83%) favored AI and 1 (17%) was mixed. For non-randomized studies, all 3 studies (100%) favored AI. By AI role, tutor/mentor systems favored AI in 5 of 6 studies (83%), feedback generators in 8 of 10 (80%), simulation-only in 1 of 5 (20%), and adaptive narratives in 1 of 1 study (100%).

Outcome-Level Direction of Effects

To further examine the direction of effects at the outcome level, Additional file 4 summarizes each reported outcome separately. Clinical reasoning/history-taking performance showed the most consistent favorable pattern, with 10 of 12 outcomes (83%) favoring AI; one outcome (8%) favored the comparator and one (8%) showed no clear difference. For knowledge/retention, all 4 outcomes (100%) favored AI. Communication/interaction also generally favored AI, with 3 of 4 outcomes (75%) favoring AI and 1 (25%) favoring the comparator. Among clinical competence/skills outcomes, 4 of 7 (57%) favored AI, whereas 2 of 7 (29%) showed no clear difference. Diagnostic accuracy was more variable, with 2 of 5 outcomes (40%) favoring AI, 2 (40%) not applicable for comparative direction, and 1 (20%) showing no clear difference.

Learner perception/satisfaction showed a similar distribution, with 2 of 5 outcomes (40%) favoring AI, 2 (40%) not applicable, and 1 (20%) mixed. All clinical reasoning/process coding outcomes (4/4, 100%) and feasibility/AI fidelity outcomes (2/2, 100%) were coded as not applicable because they were descriptive or non-comparative.

Discussion

Summary of Principal Findings

This review included 16 studies evaluating AI-enhanced virtual patients for clinical reasoning training. Overall, the findings point to enhanced clinical reasoning, with 8 of 10 studies with an external comparator reporting favorable outcomes associated with AI-VPs. Among the outcome domains, the clearest support was observed for clinical reasoning and history-taking performance, where 7 of 8 studies favored AI. Knowledge or retention outcomes favored AI in 3 of 3 studies, although this domain was assessed in fewer studies. Interventions that used AI to provide feedback or tutor/mentor support also appeared more consistently favorable, with 8 of 10 feedback-based studies and 5 of 6 tutor/mentor studies favoring AI. This suggests that the educational value of AI-VPs may depend not only on simulating patient encounters, but also on how actively the system supports learners' reasoning.

Interpretation of Overall Direction of Effects

This review suggests a promising overall signal for the promising potential of AI-enhanced virtual patients in supporting clinical reasoning, although this signal should be interpreted cautiously. This finding is consistent with broader literature on virtual patients in clinical reasoning education, which has reported potential benefits while also emphasizing heterogeneity in outcome measures and study designs (10). The present review extends this earlier VP

evidence into an AI-enhanced context, where AI may add conversational interaction, adaptive case presentation, or individualized feedback (35). The positive direction of effects is educationally plausible, as effective simulation-based learning often depends on repeated practice, feedback, and structured learning opportunities (36). Favorable findings were also reported in a continuing medical education context, suggesting that the potential relevance of AI-supported VP training may extend beyond undergraduate learners (27). However, the evidence was not uniformly favorable. Some active-comparator studies did not demonstrate clear superiority of AI-VPs (22), and in one crossover comparison, diagnostic accuracy did not differ significantly between ChatGPT-based simulation and real-patient interaction (34). Therefore, the observed favorable direction should not be interpreted as strong causal evidence as the included studies varied substantially in AI role, intervention design, feedback provision, comparator type, and outcome measurement, which limits direct comparison across studies. Overall, current evidence supports AI-VPs as supplementary tools for structured clinical reasoning practice rather than replacements for real or supervised clinical encounters (34). Stronger studies with clearer comparators, standardized clinical reasoning outcomes, and longer follow-up are needed before firm conclusions can be drawn about effectiveness and transferability (37).

Effects Across Clinical Reasoning Domains

This review examined whether the strength of evidence differed across clinical reasoning domains. The clearest positive direction of effects was observed for clinical reasoning and history-taking performance. This finding is educationally meaningful because history taking is one of the earliest stages at which clinical reasoning becomes observable: learners must select relevant cues, pursue diagnostic hypotheses, and organize patient information (38). AI-VPs may be particularly suited to this domain because they can provide repeated, interactive practice with

simulated clinical encounters(24). Moreover, history taking should not be interpreted merely as conversational fluency; it may also reflect deeper cognitive processes involved in gathering and structuring information for diagnostic problem representation(39).

Knowledge and retention outcomes also favored AI-VPs, although this finding was based on fewer studies. These outcomes are relevant because organized clinical knowledge and illness-script formation contribute to the development of clinical reasoning expertise (40). However, knowledge gains should be distinguished from performance-based reasoning outcomes, since clinical reasoning is a multidimensional construct assessed through different methods and at different levels of performance (5).

Evidence for diagnostic accuracy and broader clinical competence was more variable. Diagnostic accuracy is clinically important, but final-answer accuracy alone may not capture the quality of the reasoning process leading to that answer (41). Consistent with this caution, one crossover study found no significant difference in diagnostic accuracy between ChatGPT-based simulation and real-patient interaction (34). Learner perceptions, satisfaction, and usability were also assessed in several included studies; however, these outcomes should be interpreted as implementation-related measures rather than direct evidence of improved clinical reasoning (5).

Overall, these domain-level findings should be interpreted cautiously because outcome measures varied substantially across studies (10). Future studies should use clearer, standardized, and domain-specific measures that align with the targeted component of clinical reasoning (5). In addition, the quality of studies should also be considered when interpreting the results as a simple percentage cannot present the true direction of effects because there may be more or less methodologically robust studies in a given percentage of studies. One observed pattern was that

studies meeting more methodological criteria in MMAT frequently assessed clinical reasoning/history-taking or clinical competence/skills outcomes.

The Role of AI in Virtual Patient Design

The findings of this review conclude that the educational contribution of AI-enhanced virtual patients may depend on the role that AI plays within the instructional design. In our synthesis, AI appeared to provide a stronger signal when it functioned as a feedback generator or tutor/mentor, rather than when it was used only to simulate patient interaction. AI as a simulated patient may still be educationally valuable because it can provide repeated, low-risk practice, case exposure, and opportunities to develop history-taking and information-gathering skills (10, 24). However, simulation alone should not be assumed to outperform human interaction, as AI chatbot simulation was not clearly superior to peer role-play in at least one active-comparator study (22). This suggests that conversational realism may be insufficient unless the VP encounter is linked to reflection, feedback, or instructional scaffolding (19, 22). However, interpretation of these results should be done cautiously as one study could have several AI-roles and we could not separate the effects of each AI-role in one study. Thus, the overall direction of effects may be due to a certain AI-role in a study but be attributed to all of the AI-roles utilized in that study. Among the AI roles examined, feedback generation appeared particularly important for clinical reasoning education. Studies comparing feedback-enhanced AI-VPs with otherwise similar VP experiences without AI-generated feedback reported more favorable learner outcomes (19, 26). This may reflect the nature of clinical reasoning learning, where learners need to evaluate not only the final diagnostic conclusion but also the reasoning pathway that led to it (11) and also highlights the importance of feedback in medical education which is known to improve critical thinking process of medical students (42), a critical component of clinical reasoning skill. By

helping learners identify missed cues, incomplete questioning strategies, or weaknesses in diagnostic justification, feedback may transform VP interaction from simple case exposure into a more deliberate learning process. Nevertheless, the educational value of AI-generated feedback depends on its accuracy, transparency, and alignment with expert standards; therefore, such feedback should be validated and supervised rather than assumed to be reliable (43).

AI tutoring was another promising role identified in the included studies. The stronger signal for tutor/mentor functions is consistent with broader evidence that generative AI can support guided practice and inquiry in health professions education (44). One plausible explanation is that AI tutors may scaffold the clinical reasoning process by prompting learners to attend to relevant cues, organize information, and move through the case in a more structured way, rather than leaving them to navigate the encounter without guidance. However, evidence from AI tutoring in other simulation contexts should be interpreted cautiously when applied to VP-based clinical reasoning because the learning tasks are not identical (45).

Adaptive narrative or case-generation AI also represents a potentially useful design approach, although the available evidence remains preliminary. In this review, only one included study clearly assessed this role, but it found that adaptive AI-generated narratives produced more favorable diagnostic reasoning-related outcomes than static narratives (20). Adaptive narrative AI may support reasoning by varying case information, contextual cues, emotional tone, patient characteristics, or case complexity, thereby requiring learners to practice cue recognition and diagnostic reasoning in a more dynamic environment than static case formats (20). However, this mechanism remains insufficiently tested, and current findings should be treated as hypothesis-generating rather than definitive.

Taken together, the current literature suggests that although AI is now used in many ways in medical education and even AI itself is creating clinical guidelines (46), simply adding AI to an educational or clinical reasoning task may be insufficient without purposeful instructional design (47). A practical implication is that AI-VP design should begin with the intended reasoning outcome rather than with the technology itself: simulated-patient functions may support practice and exposure, tutor/mentor functions may support scaffolding, feedback functions may support correction and reflection, and adaptive narratives may support case variation. Accordingly, AI-VPs are best understood as complementary tools for structured clinical reasoning practice, rather than substitutes for human-supervised clinical learning (22, 47).

When interpreting the results of different AI usages in virtual patients some descriptive patterns are seen, for example 80% of AI-generated feedback studies favored AI compared to 33% of studies without feedback or 83% of tutor/mentor systems favored AI. However, this should be counted as only exploratory patterns and not as definitive direction of effects or superiority of these types of AI implementations because these studies are based on small, non-randomised groupings of heterogeneous studies.

Strengths and Limitations of This Review

This review has several strengths. It addresses a focused and timely question by examining whether AI-enhanced virtual patients can support clinical reasoning training among healthcare students and professionals. By concentrating specifically on AI-enhanced VPs and clinical reasoning-related outcomes, the review provides a more targeted contribution than broader reviews of virtual patients or digital simulation. The review was conducted systematically, with an a priori protocol, PRISMA-guided reporting, a broad search strategy, independent screening, standardized data extraction, and MMAT-based quality appraisal. Another strength is that the

synthesis went beyond reporting whether studies favored AI. We also categorized the evidence by outcome domain, AI role, feedback presence, intervention format, and comparator type. This helped identify where the signal of benefit appeared more consistent, particularly for clinical reasoning and history-taking outcomes and for systems that used AI as a feedback generator or tutor/mentor. The inclusion of randomized controlled trials also strengthens the evidence base, although their findings were interpreted cautiously alongside non-randomized, descriptive, and mixed-methods studies.

This review also has limitations. A meta-analysis was not feasible because the included studies differed substantially in design, population, AI-VP format, intervention dose, comparator condition, and outcome measurement. Therefore, the direction-of-effect synthesis should be understood as a structured summary of reported findings rather than a pooled estimate of effectiveness. Because this type of synthesis does not account for factors such as effect magnitude, methodological quality, or sample size and all studies equally contribute to the overall percentage provided. Some included studies had no comparator group, while others used pre-post designs, historical controls, traditional teaching, role-play, standardized patients, or non-AI virtual patients. This made it difficult to isolate the added value of AI itself from the broader educational value of VP-based simulation. In addition, categorizing studies into fixed outcome domains or AI roles may have simplified complex interventions, especially when the AI component, feedback mechanism, or adaptation process was incompletely described. The evidence bases also included small samples, short or unclear intervention durations, limited follow-up, and few assessments of retention or transfer to real clinical practice. Most studies were conducted among students, with fewer studies involving residents or practicing clinicians, which may limit generalizability. Methodological concerns identified through MMAT included

incomplete reporting of randomization, allocation concealment, assessor blinding, intervention adherence, confounding, representativeness, and selection or nonresponse bias. Also, we only used three databases of PubMed/MEDLINE, Scopus, and ERIC and only included English results; this may have resulted in potentially missed studies such as non-English reports or articles indexed in other databases (i.e. psychological databases or interdisciplinary literature). Finally, only English-language studies were included, some full texts were unavailable, and reporting of AI model details, prompting strategies, feedback logic, and safety monitoring was often limited. These limitations mean that the findings should be interpreted as promising but still preliminary.

Conclusions

To conclude, this systematic review suggests that AI-enhanced virtual patients have promising potential to support clinical reasoning education. The most consistent direction of effects were observed for history-taking and clinical reasoning performance, particularly in interventions where AI was used to provide feedback or tutor/mentor support. However, the current evidence remains early, heterogeneous, and methodologically variable. Further well-designed controlled studies with standardized clinical reasoning outcomes, transparent reporting of AI components, and longer follow-up are needed before stronger conclusions can be drawn about effectiveness, transferability, and implementation in health professions education.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All data generated or analysed during this study are included in this published article and its supplementary information files.

Competing interests

The authors declare that they have no competing interests.

Funding

This study received no funding.

Authors' contributions

S.M and Z.K contributed to conception and design of the work, acquisition and analysis, interpretation of data, drafting, and revising the work. both authors reviewed the final manuscript and approved it.

Acknowledgements

Not applicable.

Clinical trial number

Not applicable.

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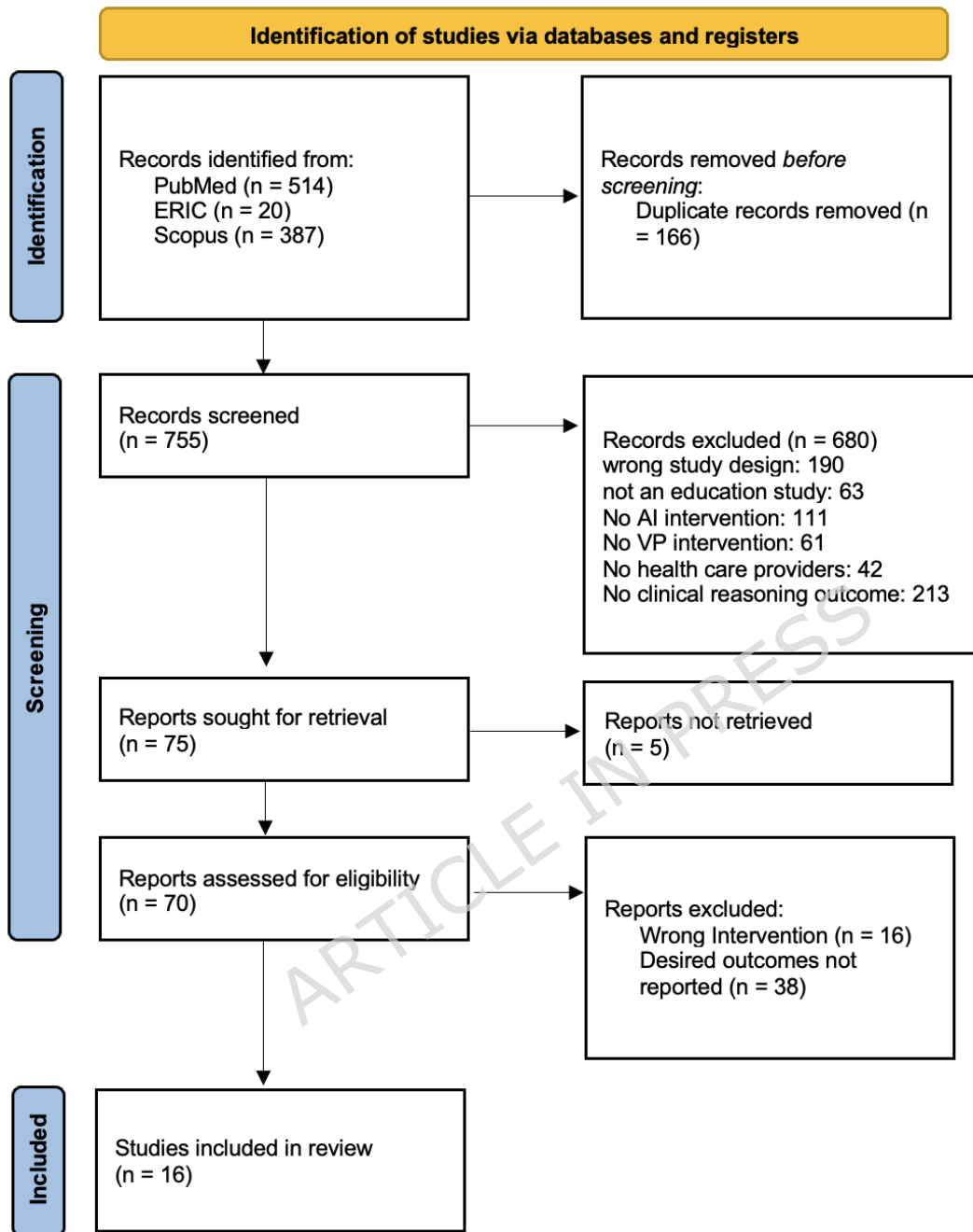
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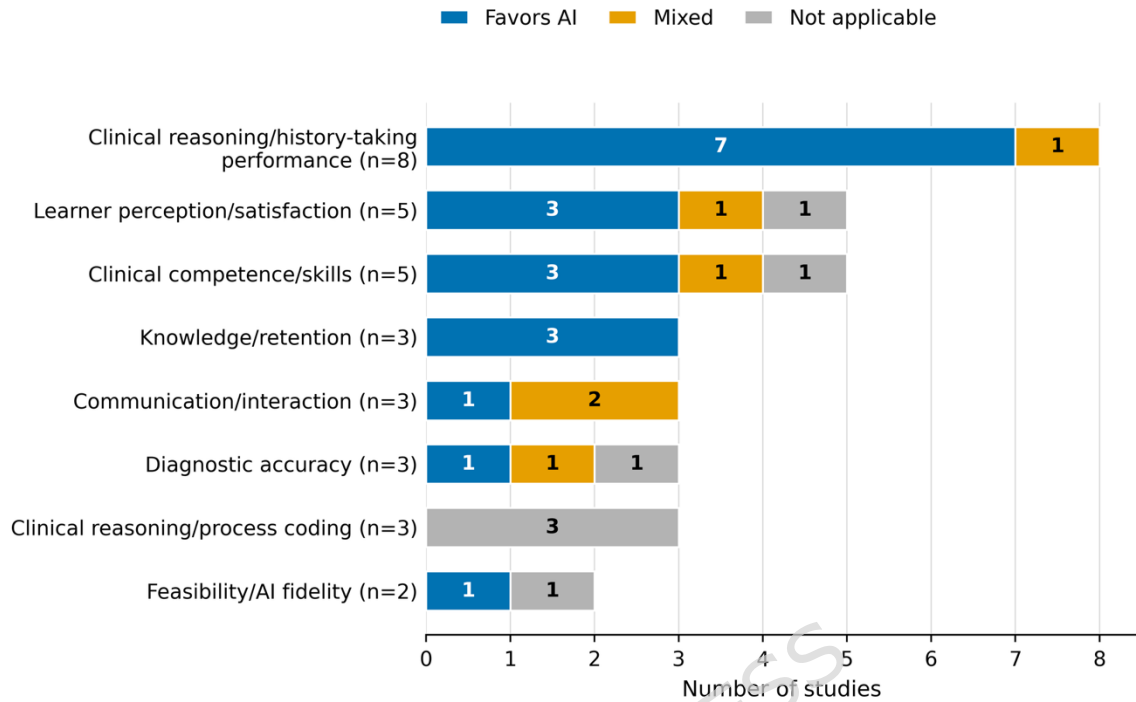
Figures

Figure 1. PRISMA flow diagram.



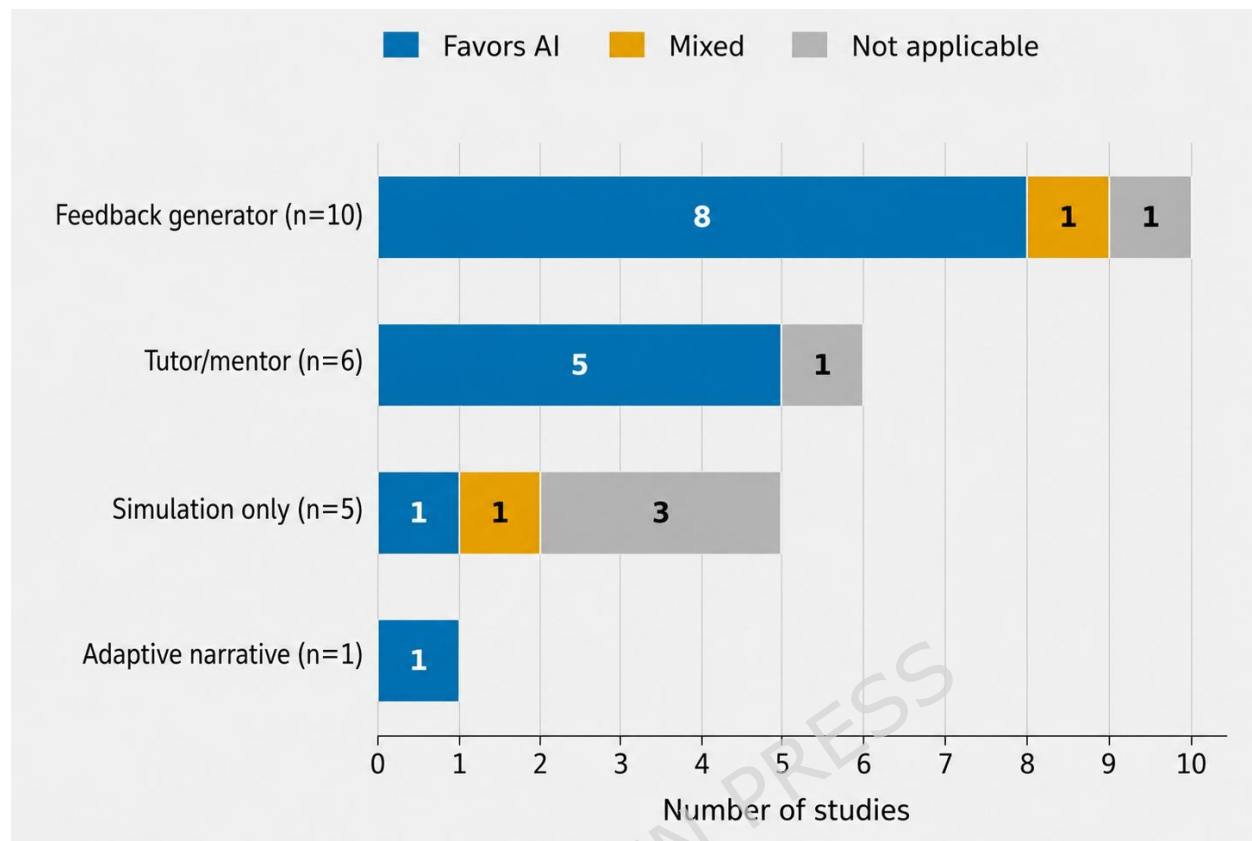
Number of records identified from each database, primary, and secondary screening process is illustrated alongside the reason of exclusion in each stage.

Figure 2. Direction of effects by outcome-domain



Overall direction of effects of studies in each outcome domain is shown. The figure presents total number of studies assessing each outcome domain and also how many of them were in favor of AI, had mixed results, or were not applicable for overall direction of effects decision.

Figure 3. Direction of effects by AI role



Overall direction of effects of studies by each AI role in VP design is shown. The figure presents total number of studies utilizing each AI role and also how many of them were in favor of AI, had mixed results, or were not applicable for overall direction of effects decision.

Additional Files

Additional file 1

.docx file

Definition of Codes Used in Categorizing Studies

Detailed definition of each category of intervention, AI-role, and outcome domain is provided.

Additional file 2

.docx file

Full Search Strategy of Each Database

The exact search string used in each database is provided in this file.

Additional file 3

.docx file

Full tables of MMAT quality assessment

Complete tables of quality assessment for each study design are provided in this file.

Additional file 4

.docx file

Outcome-Level Direction of Effects Across Included

The direction of effect and measuring tool for each outcome in every separate study is discussed in this file.

Tables

Table 1. Study characteristics and population

Study (Author, Year)	Study design	Sample size	Population	Age
Brügge et al., 2024 (19)	RCT	n=21	Medical students; mean semester 4	Mean 22.10 years
Chen et al., 2026 (31)	Descriptive/validation/observational	n=210	Second-year medical students; preclinical medicine	Not reported
Díaz, 2026 (20)	Mixed-methods pilot RCT	n=46	Second-year undergraduate nursing students	Not reported
Furlan et al., 2021 (28)	Pre-post self-comparison	n=15	Undergraduate medical students	Not reported
He et al., 2026 (29)	Pre-post self-comparison	n=28	Obstetrics and gynecology residents	Mean 25.7±1.8 years; range 23–30
Holderried et al., 2026 (32)	Descriptive/feasibility	n=148;	Medical students, mostly sixth semester	Mean 22.71 years
Ji et al., 2026 (21)	RCT	n=124; intervention n=62, control n=62	Breast oncology residents	Intervention 27.8±2.1 years; control 28.1±2.3 years
Lee et al., 2025 (22)	Pilot RCT	n=19; AI chatbot n=9, peer role-play n=10	Final-year Korean medicine students	Not reported
Liu et al., 2026 (23)	RCT; mixed-methods	n=60; experimental n=30, control n=30	Nursing students	Not reported
Öncü et al., 2025 (33)	Mixed-methods descriptive	n=21	Sixth-year medical students / intern physicians	Not reported
Rodrigues-Pereira et al., 2025 (34)	Blinded cross-sectional crossover	n=30	Undergraduate dental students	Not reported
Wang et al., 2025 (24)	RCT	n=56; GPT n=28, control n=28	Medical students before intensive clerkship	Not reported
Yamamoto et al., 2024 (25)	Nonrandomized controlled	n=145; AI n=35, historical control n=110	Fourth-year medical students	AI group median 22 years; control median 23 years
Borg et al., 2026 (26)	Quasi-experimental nonrandomized controlled	n=115; feedback n=61, control n=54	Sixth-semester medical students in rheumatology rotation	Feedback group 26.0±5.9 years; control 26.8±5.3 years
Jarrell et al., 2015 (30)	Observational correlational/cluster	n=30	First- and second-year medical students	Mean 23 years; SD 2.60

Study (Author, Year)	Study design	Sample size	Population	Age
	analysis			
McFadden & Crim, 2016 (27)	Nonrandomized comparative CME study	n=68; treatment n=41, control n=27	Practicing primary care providers	Not reported

RCT, randomized controlled trial; CME, continuing medical education; AI, artificial intelligence; GPT, generative pre-trained transformer.

Table 2. Intervention features and comparator conditions

Study (Author, Year)	Intervention format	AI role	Clinical domain	Intervention dose	Feedback	Comparator group
Brügge et al., 2024 (19)	Text-based chatbot VP	Simulation; Feedback	History taking / clinical decision-making	4 training sessions	Yes; AI-generated structured feedback	AI-simulated patient conversation without feedback
Chen et al., 2026 (31)	Text-based chatbot VP	Simulation	Chest pain history taking	5 VP cases over 5 weeks	No evaluated AI feedback	No comparator
Díaz, 2026 (20)	Text-based adaptive narrative simulation	Simulation; Adaptive Narrative	Nursing diagnostic reasoning	Single 90-minute session	No evaluated AI feedback	Static narrative control
Furlan et al., 2021 (28)	Text-based natural-language VP	Tutor/Mentor; Feedback	Diagnostic reasoning	2-hour test; prior simulated cases in preceding weeks	Yes; real-time step-by-step feedback	Pre-simulation baseline
He et al., 2026 (29)	Case simulation platform	Tutor/Mentor; Feedback	Obstetrics and gynecology	12 months	Yes; AI-guided prompts and competency feedback	Pre-training baseline
Holderried et al., 2026 (32)	Text-based chatbot VP	Simulation	Mental health / depression and suicidality	Single VP interaction	No evaluated AI feedback	No comparator
Ji et al., 2026 (21)	MR simulation with VP-AI mentor	Simulation; Tutor/Mentor; Feedback	Breast oncology	3-month rotation	Yes; immediate error feedback and analytics-supported feedback	Conventional residency training
Lee et al., 2025 (22)	Text-based chatbot VP	Simulation; Feedback	Korean medicine OSCE preparation	30-minute physical examination practice plus 1-hour interview session	Yes; automated feedback	Peer role-play under tutor supervision
Liu et al., 2026 (23)	Text-based chatbot VP	Simulation	Nursing clinical thinking	Not reported	No evaluated AI feedback	Conventional/traditional training
Öncü et al., 2025 (33)	Text-based chatbot VP	Simulation	Clinical case management	Case interaction; 5-minute limit noted	No evaluated AI feedback	No comparator
Rodrigues-Pereira et al., 2025 (34)	Text-based chatbot / case simulation	Simulation	TMD / orofacial pain	Not reported	No evaluated AI feedback	Standardized real patient

Study (Author, Year)	Intervention format	AI role	Clinical domain	Intervention dose	Feedback	Comparator group
Wang et al., 2025 (24)	Text/voice-enabled chatbot VP	Simulation; Feedback	History taking	Three 30-minute sessions/week for 4 weeks	Yes; automated written rubric-based feedback	Traditional role-play with clinical-skills instructors
Yamamoto et al., 2024 (25)	Text-based chatbot VP	Simulation; Feedback; Tutor/Mentor	Medical interviewing	Not reported as fixed duration	Yes; feedback from simulated patient/instructor program	Historical/traditional practice control
Borg et al., 2026 (26)	Social robotic VP with virtual cases	Simulation; Feedback	Rheumatology history taking	9 VP cases over a 6-week rotation	Yes; LLM-generated structured feedback	Same VP cases without AI-generated feedback
Jarrell et al., 2015 (30)	Intelligent tutoring VP	Simulation; Feedback; Tutor/Mentor	Diagnostic reasoning / medical problem-solving	Not reported	Yes; individualized performance feedback	No comparator
McFadden & Crim, 2016 (27)	Online simulation/tutor CME	Simulation; Feedback; Tutor/Mentor	Joint pain / rheumatology	Online duration not clearly reported; pre/post tests up to 30 minutes	Yes; AI-driven feedback for diagnostic errors	Live expert-led multimedia CME

Abbreviations: AI, artificial intelligence; CME, continuing medical education; LLM, large language model; MR, mixed reality;

OSCE, objective structured clinical examination; TMD, temporomandibular disorders; VP, virtual patient.

Table 3. Summary of findings

Study (Author, Year)	Outcome Domain	Measurement Tool	Direction of Effect	Summary of Findings
Brügge et al., 2024 (19)	Clinical reasoning/history-taking performance	CRI-HTI, two-rater assessment; repeated-measures ANOVA	Favors AI	Structured LLM feedback improved overall CRI-HTI and two clinical decision-making subdomains; focusing questions did not improve significantly.
Chen et al., 2026 (31)	Clinical reasoning/process coding	12-behavior coding scheme; interrater reliability; correlations with performance metrics	Not applicable	The coding scheme showed high interrater reliability and identified clinical reasoning behaviors associated with performance metrics; no intervention effect was tested.

Study (Author, Year)	Outcome Domain	Measurement Tool	Direction of Effect	Summary of Findings
Díaz, 2026 (20)	Clinical reasoning/history-taking performance; Learner perception/satisfaction	NIC-NOC coherence assessment; self-efficacy measure	Favors AI	Adaptive AI-generated narratives showed higher NIC-NOC coherence, and self-efficacy than static narrative cases.
Furlan et al., 2021 (28)	Knowledge/retention	22-item MCQ pre/post assessment	Favors AI	Scores increased from pre- to post-simulation, including the overall MCQ and core case-specific questions.
He et al., 2026 (29)	Clinical reasoning/history-taking performance; Knowledge/retention; Clinical competence/skills	Four-dimensional competency framework; case analysis parameters	Favors AI	Post-training scores improved across professional knowledge, clinical reasoning, clinical skills, and case-analysis indicators.
Holderried et al., 2026 (32)	Feasibility/AI fidelity; Diagnostic accuracy; Learner perception/satisfaction	Expert chat analysis; student diagnostic choices; questionnaire	Not applicable	The chatbot maintained assigned roles in most answers; descriptive diagnostic accuracy and acceptability outcomes were reported without a comparator.
Ji et al., 2026 (21)	Clinical competence/skills; knowledge/retention	Knowledge, reasoning, procedural, collaboration assessments; retention test	Favors AI	The AI-empowered intelligent teaching group outperformed conventional training across reported competence domains and knowledge retention.
Lee et al., 2025 (22)	Clinical competence/skills; Clinical reasoning/history-taking performance; Communication/interaction; Learner perception/satisfaction	Two-station OSCE checklist; post-study questionnaire	Mixed	Overall OSCE differences were not statistically significant; peer role-play tended higher for history taking, while AI tended higher for patient education and some learner-experience items.

Study (Author, Year)	Outcome Domain	Measurement Tool	Direction of Effect	Summary of Findings
Liu et al., 2026 (23)	Clinical reasoning/history-taking performance; Communication/interaction; Learner perception/satisfaction;	LCJR; SEGUE framework; interviews/satisfaction survey	Favors AI	The AI virtual patient group had higher clinical judgment and communication scores; learner perception outcomes were reported descriptively.
Öncü et al., 2025 (33)	Clinical competence/skills; Clinical reasoning/process coding;	Self-assessment survey; observation form; semi-structured interviews	Not applicable	Observed and self-assessed competencies were described, with positive correlations reported; no comparative intervention effect was tested.
Rodrigues-Pereira et al., 2025 (34)	Diagnostic accuracy; communication/interaction	Diagnostic accuracy assessment; quantitative interaction parameters; content analysis	Mixed	Diagnostic accuracy did not differ significantly; ChatGPT provided denser information, while real-patient interactions elicited more follow-up questions and reformulations.
Wang et al., 2025 (24)	Clinical reasoning/history-taking performance; Learner perception/satisfaction	Structured clinical examination/OSCE; student survey	Favors AI	GPT-based history-taking training improved structured clinical examination performance and several learner-evaluation ratings compared with role-play.
Yamamoto et al., 2024 (25)	Clinical reasoning/history-taking performance; Feasibility/AI fidelity;	Pre-CC OSCE; SBT-QA10 survey	Favors AI	The AI group had higher medical interview OSCE scores than the historical control group; no significant safety concerns were reported.
Borg et al., 2026 (26)	Clinical reasoning/history-taking performance; Clinical competence/skills	8-minute OSCE rubric; pass/fail threshold	Favors AI	AI-generated feedback improved total OSCE score, generic history score, and pass rate; other OSCE domains showed no significant differences.
Jarrell et al., 2015 (30)	Clinical reasoning/process coding	Achievement Emotions Questionnaire adaptation; BioWorld case performance; cluster analysis	Not applicable	Achievement-emotion clusters were associated with diagnostic reasoning performance; no AI intervention comparator was tested.
McFadden & Crim, 2016 (27)	Diagnostic accuracy	14 case vignettes pre/post; repeated-measures ANOVA	Favors AI	The online AI-driven simulation/tutor CME group improved significantly on diagnostic accuracy, unlike the live CME control group.

Abbreviations: AI, artificial intelligence; CME, continuing medical education; CRI-HTI, Clinical Reasoning Indicator-History Taking Inventory; LCJR, Lasater Clinical Judgment Rating Scale; LLM, large language model; OSCE, objective structured clinical examination; Pre-CC, pre-clinical clerkship; SBT-QA10, Simulation-Based Training Quality Assurance 10; SEGUE, Set the stage, Elicit information, Give information, Understand the patient perspective, End the encounter; VP, virtual patient.

Table 4. MMAT 2018 quality appraisal summary for included studies

Study (Author, Year)	Screening questions	Category of study	Main concerns
Brügge et al., 2024 (19)	Passed	2. Quantitative randomized controlled trial	2.1. Sequence generation and allocation concealment not fully reported.
Chen et al., 2026 (31)	Passed	4. Quantitative descriptive study	4.2. Representativeness beyond the study context unclear.
Díaz, 2026 (20)	Passed	5. Mixed methods study	5.5. Component-level quality details incompletely reported.
Furlan et al., 2021 (28)	Passed	3. Quantitative non-randomized study	3.1. Representativeness unclear; 3.2. Psychometric validation of the outcome measure not reported; 3.4. Confounding not controlled.
He et al., 2026 (29)	Passed	3. Quantitative non-randomized study	3.4. Time-related or cohort confounding not controlled.

Study (Author, Year)	Screening questions	Category of study	Main concerns
Holderried et al., 2026 (32)	Passed	4. Quantitative descriptive study	4.1. Sampling process unclear; 4.2. Representativeness unclear; 4.4. Nonresponse/selection bias unclear.
Ji et al., 2026 (21)	Passed	2. Quantitative randomized controlled trial	2.5. Adherence/completion of all intervention components not fully detailed.
Lee et al., 2025 (22)	Passed	2. Quantitative randomized controlled trial	2.1. Sequence generation and allocation concealment not fully described; 2.2. Baseline sex imbalance.
Liu et al., 2026 (23)	Passed	2. Quantitative randomized controlled trial	2.1. Allocation concealment not clearly reported; 2.4. Assessor blinding unclear; 2.5. Intervention adherence unclear.
Öncü et al., 2025 (33)	Passed	5. Mixed methods study	5.5. Component-level quality uncertain because of researcher-developed instruments and convenience sampling.
Rodrigues-Pereira et al., 2025 (34)	Passed	2. Quantitative randomized controlled trial	2.1. Allocation concealment not reported; 2.2. Baseline comparability by sequence not reported.
Wang et al., 2025 (24)	Passed	2. Quantitative randomized controlled trial	No major MMAT concern recorded.
Yamamoto et al., 2024 (25)	Passed	3. Quantitative non-randomized study	3.1. Historical/consenting cohorts limited representativeness; 3.3. Questionnaire completeness unclear; 3.4. Academic-year and self-selection confounding not controlled; 3.5. Exposure/adherence unclear.

Study (Author, Year)	Screening questions	Category of study	Main concerns
Borg et al., 2026 (26)	Passed	3. Quantitative non-randomized study	3.1. Representativeness not fully established; 3.4. Site/week allocation confounding not controlled beyond baseline comparisons.
Jarrell et al., 2015 (30)	Passed	4. Quantitative descriptive study	4.1. Sampling not sufficiently reported; 4.2. Representativeness unclear; 4.4. Nonresponse bias unclear.
McFadden & Crim, 2016 (27)	Passed	3. Quantitative non-randomized study	3.1. Convenience samples from different CME contexts; 3.3. Complete outcome data not fully described; 3.4. Recruitment source, delivery mode, and training-dose confounding not controlled.

Abbreviations: MMAT, Mixed Methods Appraisal Tool; CME, continuing medical education. Screening questions were considered passed when both MMAT screening items were rated Yes. No overall numerical score was calculated.